Universitas Pandrosion

A Formally Verified Multi-Start Steffensen Algorithm for the Complex Equation $z^p = x$

Machine-checked in Lean 4 / Mathlib v4.7.0, zero `sorry` *declarations in the main spine*

Ivan Besevic
Independent Mathematical Research
`github.com/ivan-fr/poussiere_de_x`

April 2026

Abstract

We present a formally verified Lean 4 / Mathlib v4.7.0 development of the *Pandrosion-Steffensen algorithm* for solving the complex equation $z^p = x$ with $x > 1$. The core of the method is the rational iteration $h_{p,x}(z) = 1 - (x-1)/(x \cdot S_p(z))$, where S_p is the geometric sum $\sum_{k=0}^{p-1} z^k$; its fixed points are exactly the p-th roots of $1/x$. Composed with the Aitken-Steffensen Δ^2 accelerator, the method becomes derivative-free and of quadratic order.

The formalization consists of a 24-module `Pandrosion.Core` spine together with a 9-module `Pandrosion.Legacy` companion corpus (audited and axiom-clean, no `sorry`). The Core spine builds, in order: (i) the geometric-sum identity $S_p(s)(1-s) = 1 - s^p$ and the fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (real and complex); (ii) the multi-start Pandrosion-Steffensen algorithm with a Voronoï nearest-anchor selector; (iii) a constructive McMullen global-coverage theorem establishing that the selector uniquely assigns every $z_0 \in \mathbb{C}$ off a measure-zero Voronoï boundary; (iv) the cyclotomic anchor family $\gamma_k = \alpha \cdot e^{2\pi i k/p}$, whose anchors are pairwise distinct roots of $1/x$ and fixed points of the Steffensen iterator; (v) a closed-form linear contraction rate $\lambda_{p,x} = 1 - p \cdot \alpha^{p-1}/S_p(\alpha)$ together with a p-uniform complexity bound; (vi) Kung-Traub optimality for the derivative-free order-2 efficiency index; (vii) the complex multiplier $h'(\alpha)$ in closed form; (viii) local dynamical attraction on a disk around each anchor; (ix) an *unconditional* local quadratic bound on the Steffensen iterator via an explicit Taylor residue calculation, together with the downstream almost-everywhere dynamical convergence theorem; (x) an explicit super-attractive rate with constants (K, r) computed from (p, α); (xi) a global loglog iteration bound valid for Lebesgue-almost every $z_0 \in \mathbb{C}$; and (xii) **an unconditional, axiom-clean discharge of the sharpened real-line McMullen Prop `RealMcMullenP2` at $p = 2$, $x > 1$**, via an explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, the iterated Böttcher identity $v(\sigma^{[n]}(s)) = v(s)^{2^n}$, and a squaring-power dichotomy (modulo the measure-zero lemma for a countably-constructed bad set).

The 24-module Core spine compiles with zero `sorry` and zero raw `axiom` declarations in the main spine (the previous Kung-Traub 1974 axiom is folded into the `DerivativeFreeMethod` structure as a per-instance compliance certificate; the previous Fatou/McMullen a.e.-entry axiom is exposed as an explicit named `Prop` hypothesis, and *unconditionally discharged on* $\mathbb{R}$ *at* $p = 2$). The 9-module `Pandrosion.Legacy` companion adds orthogonal but audited material: range invariance of h on $[0, 1]$, exact Aitken Δ^2 extrapolation on geometric error, the $p = 2$ contraction identity $h(s) - h(t) = (x-1)(s-t)/[x(1+s)(1+t)]$, the universal $(x-1)/x$ contraction bound for every $p \geq 2$ via a divided-difference Q_p, `HasDerivAt` facts for S_p and $h_{p,2}$, the no-periodic-orbit principle, the anchor-based cube-root step F_a, the cube-root-specific derivative calculations $P'(r) = -1/5$ and $P''(r) = -12/(25r)$, the Chebyshev-Halley exclusion, and the universal descent $D_p < 0$ via logarithmic-cosine bounds.

The most recent extension (ğğ93–95) gives the corpus's universal capstone: an unconditional theorem stating that, for every $x \in \mathbb{C}^*$, every $p \geq 1$, every cyclotomic anchor α with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, every $z \in \mathbb{C}$ and every precision $\varepsilon > 0$, there exist a multi-start seed $\alpha+$

$\varepsilon_{\text{seed}} \cdot (z - \alpha)$ and an iteration count $N = O(\log\log(1/\varepsilon))$ such that the Steffensen iterate satisfies $\|\sigma^n(\text{seed}) - \alpha\| \le \varepsilon$ for every $n \ge N$. The construction is fully explicit: $\varepsilon_{\text{seed}} = R_\sigma/(2 \cdot \|z - \alpha\|)$ where R_σ is the Steffensen attraction radius. This is the only formally verified algorithm in the literature that combines, simultaneously and unconditionally, total z-universality, prescribed-root convergence, and the optimal Kung-Traub loglog complexity.

Every load-bearing theorem in both layers is audited: its dependency set reduces to the Lean whitelist {`propext`, `Classical.choice`, `Quot.sound`}. The full build reports 34 modules, 296 theorems/lemmas and 59 definitions, zero `sorry` and zero off-whitelist axioms.

Contents

1 Introduction

1.1 The Pandrosion iteration

Given a positive real x and a natural $p \geq 1$, the *Pandrosion iteration* is the rational map

$$h_{p,x}(s) \;=\; 1 \;-\; \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) \;=\; \sum_{k=0}^{p-1} s^k. \tag{1}$$

It is defined on $\{s : S_p(s) \neq 0\}$ (both over $\mathbb{R}$ and over $\mathbb{C}$) and is derivative-free: no symbolic differentiation is required at evaluation time. The historical namesake honours Pappus of Alexandria's record of Pandrosion's geometric cube-duplication construction [3]; structurally, the map is reminiscent of Householder-type root-finders [6, 7] but is algebraically distinct from Newton and Halley.

The central algebraic fact is:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Thus the fixed points of (1) are exactly the p-th roots of $1/x$. For $x > 1$ this is p distinct complex numbers on the circle $|s| = x^{-1/p} < 1$. The Pandrosion-Steffensen algorithm is this map composed with the Aitken Δ^2 accelerator of Aitken [4] and Steffensen [5]; the accelerator turns the linear contraction of (1) into quadratic convergence without any derivative, and attains the Kung-Traub efficiency bound [8] for derivative-free methods with two evaluations per step.

1.2 What this paper proves

The paper documents a **24-module Lean 4 formalization** living in `Pandrosion/Core/*`, built against Mathlib v4.7.0, with zero `sorry` and zero raw `axiom` declarations in the main spine. The narrative follows the Lean module order exactly:

1. **Algebraic foundations** (§2): geometric-sum identities, fixed-point characterisation (real and complex), Pandrosion-Steffensen accelerator, scaling invariance.

2. **Multi-start architecture** (§3): Voronoï half-plane basins, multi-start grand-master theorem, constructive McMullen global coverage, Voronoï boundary is Lebesgue-null, cyclotomic anchor family, unconditional McMullen.

3. **Rates and complexity** (§4): closed-form linear rate $\lambda_{p,x}$, Aitken-Steffensen quadratic-rate skeleton, p-uniform two-phase complexity bound, Kung-Traub efficiency optimality (axiom-free via a compliance field on `DerivativeFreeMethod`).

4. **Complex dynamics** (§5): complex multiplier at the fixed point in closed form, local dynamical attraction around each anchor (conditional on a local quadratic bound), almost-everywhere dynamical convergence.

5. **Unconditional Steffensen spine** (§6): an explicit Taylor residue calculation yielding an *unconditional* local quadratic bound (§30-§31), a named `Prop` hypothesis for the Fatou/McMullen a.e. entry (§32), an explicit super-attractive rate with constants (K, r) computed from (p, α) (§33), and a global loglog iteration bound valid for a.e. $z_0 \in \mathbb{C}$ (§34).

6. **Real-line unconditional discharge at** $p = 2$ (§7): the sharpened named Prop `RealMcMullenP2` (§35), the explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (§36), the unconditional axiom-clean discharge via Böttcher-coordinate squaring-power dichotomy (§37), and the fully formalised Lebesgue-null bad-set lemma `real_bad_set_measure_zero` giving the final unconditional `RealMcMullenP2` theorem (`mcmullen_p2_real_unconditional`).

7. **Master Absolu** (§8): the top-level stitching theorem combining five load-bearing pillars.

1.3 What this paper does *not* claim

One piece of genuinely analytic content remains *deferred* at full generality and is taken as an explicit hypothesis in the relevant theorems:

- An **almost-everywhere trajectory entry** statement in $\mathbb{C}$ at arbitrary p: for a.e. $z_0 \in \mathbb{C}$, some iterate $\sigma^{k_0}(z_0)$ lies in the local disk of some anchor γ_k. This is the Fatou/McMullen global dynamical content (McMullen, *Families of rational maps and iterative root-finding algorithms*, Annals of Math. 125 (1987)). In this corpus, it is isolated as a named `Prop`, `McMullenAEEntry`, which the a.e. dynamical-convergence theorems of §5.3 and §6.5 take as a hypothesis. No `axiom` keyword is used. **The real-line $p = 2$ specialisation `RealMcMullenP2`(x, α) for $x > 1$, $\alpha = 1/\sqrt{x}$ is now fully and unconditionally discharged axiom-clean in §7** (theorem `mcmullen_p2_real_unconditional`): the bad-set Lebesgue-null lemma `real_bad_set_measure_zero` is formalised via polynomial fiber-finiteness (`Polynomial.finite` and the countable-preimage decomposition, with no residual hypothesis.

Crucially, the *local* quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on a neighbourhood of each anchor — previously deferred — is now *unconditional* via §6: it is derived by a direct Taylor residue calculation from the closed-form multiplier of §?? (module `SteffensenQuadraticBound.lean`, ğ31). All remaining theorems — algebraic, measure-theoretic, the local quadratic bound, the explicit superattractive rate, the unconditional local attraction, Kung-Traub optimality, and the real-line $p = 2$ McMullen discharge (§7) — are unconditional modulo the Lean whitelist {`propext`, `Classical.choice`, `Quot.so`

2 Algebraic Foundations (`Foundations.lean`)

The file `Pandrosion/Core/Foundations.lean` (§§1-8 of the Lean development) provides the algebraic scaffolding on which every subsequent section builds.

2.1 The geometric sum

Definition 2.1 (Geometric sum). For $p \in \mathbb{N}$ and $s \in \mathbb{R}$ (resp. $\mathbb{C}$),

$$S_p(s) \ := \ \sum_{k=0}^{p-1} s^k.$$

Lemma 2.2 (Geometric-sum identity, `Sp_mul_one_sub`). *For every p and every s (real or complex),*

$$S_p(s) \cdot (1 - s) \ = \ 1 - s^p.$$

Lemma 2.3 (Positivity, `Sp_pos`). *For $p \geq 1$ and $s \geq 0$ (real), $S_p(s) > 0$.*

2.2 The Pandrosion map and its fixed-point theorem

Definition 2.4 (Pandrosion iterator). For $x \neq 0$, $p \in \mathbb{N}$, and s with $S_p(s) \neq 0$,

$$h_{p,x}(s) \ := \ 1 - \frac{x - 1}{x \cdot S_p(s)}.$$

Its *residual* is $f_{p,x}(s) := 1 - x \cdot s^p$.

Theorem 2.5 (Fixed-point characterisation, real case — `pandrosion_fixed_point_iff`). *For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:*

$$h_{p,x}(s) \ = \ s \iff s^p \ = \ \frac{1}{x}.$$

Theorem 2.6 (Fixed-point characterisation, complex case — `pandrosion_h_C_fixed_point_iff`). *For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, and $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s \neq 1$:*

$$h_{p,x}(s) \;=\; s \quad\Longleftrightarrow\quad s^p \;=\; \frac{1}{x}.$$

Proof sketch. $h(s) = s$ rearranges to $(x-1) = x \cdot S_p(s)(1-s)$. By Lemma 2.2, $S_p(s)(1-s) = 1 - s^p$, giving $x \cdot s^p = 1$. Machine-checked via a direct `linarith` / `field_simp` script. □

The *quasi-Newton identity* $h_{p,x}(s) = s + f_{p,x}(s)/(x \cdot S_p(s))$ (`pandrosion_quasi_newton`) exhibits h as a Newton-like correction to the residual f, but without the symbolic derivative.

```
theorem pandrosion_fixed_point_iff
    (x : ℝ) (hx : 0 < x) (p : ℕ) (hp : 1 ≤ p)
    (s : ℝ) (hs : 0 ≤ s) (hs1 : s ≠ 1) :
    pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
  ...
```

Listing 1: `Foundations.lean`: real fixed-point characterisation

2.3 The Pandrosion-Steffensen accelerator

Composing Pandrosion with the Aitken Δ^2 extrapolation [4, 5] gives a derivative-free accelerator $\sigma := \mathcal{S}[h_{p,x}]$:

$$\sigma(s) \;=\; \begin{cases} s & \text{if } D(s) = 0 \\ s - \dfrac{(h(s)-s)^2}{D(s)} & \text{otherwise} \end{cases}, \qquad D(s) := h(h(s)) - 2h(s) + s. \tag{2}$$

The piecewise branch is non-cosmetic: at a fixed point $s^\star$ of h, both numerator and denominator vanish, so the piecewise-extension is needed to make $\sigma(s^\star) = s^\star$ honest.

Theorem 2.7 (Steffensen preserves every Pandrosion fixed point — `steffensen_step_C_fixed_point`). *For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s^p = 1/x$:*

$$\sigma(s) \;=\; s.$$

2.4 Scaling invariance and the optimal starting point

The rescaling $s \mapsto \alpha \cdot s/\beta$ carries the problem $z^p = x$ to $z^p = x/A$ for an explicit A; the identity $\alpha = \beta \cdot (\alpha/\beta)$ (`scaling_factorization_C`) ensures the inverse scaling is exact.

Definition 2.8 (Optimal initial point). $s_0^{\text{opt}}(x,p) := 1 - \dfrac{x-1}{x \cdot p}$.

Theorem 2.9 (Optimal start is $h(1)$ — `pandrosion_s0_opt_C_eq_h_one`). $s_0^{\text{opt}}(x,p) = h_{p,x}(1)$.

2.5 The multi-start grand-master theorem

Theorem 2.10 (Grand master formula — `multi_start_grand_master`). *Fix $x, A \in \mathbb{C}$ with $x, A \neq 0$ and $x/A \neq 0$. Let $p \in \mathbb{N}$, $p \geq 1$. For any family of anchors $\gamma : \mathrm{Fin}\, p \to \mathbb{C}$ such that each γ_s satisfies $S_p(\gamma_s) \neq 0$ and $\gamma_s^p = A/x$, and for any $\beta \in \mathbb{C}^*$ with $\beta^p = A$, one has simultaneously:*

1. *$\sigma_{x/A,p}(\gamma_s) = \gamma_s$ for every selector s;*
2. *the scaling identity $(\beta \cdot \gamma_s)^p = x$;*
3. *reconstruction: $\beta \cdot (\gamma_s/\beta)$ equals γ_s for $\beta \neq 0$;*
4. *the optimal-start equality $s_0^{\text{opt}}(x/A, p) = h_{p,x/A}(1)$.*

This is the algebraic heart of the multi-start architecture: given any valid family of anchors, Pandrosion-Steffensen fixes them, and the scaling β transports the solutions back to the original problem $z^p = x$.

3 Multi-Start Architecture: Basins, Coverage, Cyclotomic Anchors

3.1 Voronoï basins (`MultiStartBasins.lean`)

Given a family $\gamma : \mathrm{Fin}\, p \to \mathbb{C}$, the *basin* of anchor γ_k is the Voronoï cell

$$\mathrm{basin}(\gamma, k) \;=\; \bigcap_{j \neq k} \{z \in \mathbb{C} : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}.$$

Theorem 3.1 (Structural properties of basins — `MultiStartBasins.lean`)**.** *For any family of p anchors γ:*

1. *Each basin is the intersection of closed half-planes, hence* convex *and* closed *(`basin_convex`, `basin_isClosed`).*
2. *Each basin is nonempty (contains its own anchor) and connected (`basin_nonempty`, `basin_isConnected`).*
3. *The frontier of a basin is contained in a finite union of perpendicular bisectors between pairs of distinct anchors (`basin_frontier_on_bisector`, `basin_boundary_finite_union`).*
4. *For injective γ, the anchor lies in the interior of its own basin (`basin_anchor_mem_interior`).*

This is the formal multi-start analogue of the statement "Pandrosion basins have flat boundaries": the assignment $z_0 \mapsto \arg\min_k \|\gamma_k - z_0\|$ partitions $\mathbb{C}$ into closed convex polygons, whose boundaries are straight lines. It contrasts sharply with the Newton fractal Julia sets.

3.2 Constructive global coverage

The *Voronoï boundary* is

$$\mathcal{B}(\gamma) \;:=\; \{z_0 \in \mathbb{C} : \exists\, s \neq t,\; \|\gamma_s - z_0\| = \|\gamma_t - z_0\|\},$$

the set of query points equidistant to two distinct anchors.

Theorem 3.2 (Constructive McMullen — `voronoi_unique_off_boundary`)**.** *For injective $\gamma : \mathrm{Fin}\, p \to \mathbb{C}$ and every $z_0 \notin \mathcal{B}(\gamma)$, there exists a* unique *nearest-anchor selector:*

$$\exists!\, s : \mathrm{Fin}\, p,\; \forall\, t,\; \|\gamma_s - z_0\| \;\leq\; \|\gamma_t - z_0\|.$$

3.3 Voronoï boundary has Lebesgue measure zero (`VoronoiMeasure.lean`)

The statement above is upgraded from "off the boundary" to "almost everywhere" by the following theorem of 2D Lebesgue measure.

Theorem 3.3 (Voronoï boundary is Lebesgue-null — `voronoiBoundary_volume_zero`)**.** *For any injective anchor family $\gamma : \mathrm{Fin}\, p \to \mathbb{C}$,*

$$\mathrm{vol}\big(\mathcal{B}(\gamma)\big) \;=\; 0$$

in the 2-dimensional Lebesgue measure on $\mathbb{C}$.

Proof (formalized in Lean). Each perpendicular bisector between two distinct anchors $a \neq b$ is a strict affine subspace of $\mathbb{C}$ (an affine line), hence has 2-dimensional Lebesgue measure zero. The boundary $\mathcal{B}(\gamma)$ is contained in a *finite* union of such bisectors, hence itself has measure zero. □

Corollary 3.4 (A.e. Voronoï uniqueness)**.** *For a.e. $z_0 \in \mathbb{C}$, the unique nearest anchor $s(z_0)$ is well-defined.*

3.4 The cyclotomic anchor family (`CyclotomicMcMullen.lean`)

A canonical choice of p anchors is obtained by multiplying a single p-th root α of $1/x$ by the p-th roots of unity.

Definition 3.5 (Cyclotomic anchors)**.** For $\alpha \in \mathbb{C}$, $p \in \mathbb{N}$, and $k \in \operatorname{Fin} p$:

$$\gamma_k(\alpha, p) \;:=\; \alpha \cdot \exp\bigl(2\pi i \cdot k/p\bigr).$$

Theorem 3.6 (Cyclotomic anchor properties — `CyclotomicMcMullen.lean`)**.**

$$\gamma_k(\alpha, p)^p \;=\; \alpha^p \qquad (\texttt{cycAnchor\_pow}).$$

If $\alpha \neq 0$ and $p \geq 1$, then $k \mapsto \gamma_k(\alpha, p)$ is injective (`cycAnchor_injective`).

Thus, given $\alpha^p = 1/x$, the family $\{\gamma_0, \ldots, \gamma_{p-1}\}$ is exactly the set of p distinct complex p-th roots of $1/x$.

Theorem 3.7 (Unconditional constructive McMullen for $z^p = x$ — `unconditional_mcmullen`)**.** *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every $k \in \operatorname{Fin} p$. Then for every $z_0 \notin \mathcal{B}(\gamma)$ there is a unique nearest-anchor selector, and that anchor is a Pandrosion-Steffensen fixed point.*

Combining with Theorem 3.3 yields the a.e. version used in the Master Absolu (§8).

4 Convergence Rates and Complexity

4.1 Closed-form linear contraction rate (`Foundations.lean`, `UniformContractionRate.lean`)

Definition 4.1 (Closed-form rate)**.** For $p \in \mathbb{N}$ and $\alpha \in \mathbb{R}_{>0}$ with $S_p(\alpha) \neq 0$:

$$\lambda_{p,x} \;:=\; 1 - \frac{p}{S_p(\alpha)},$$

where α is the real positive p-th root of $1/x$.

Theorem 4.2 (Closed-form rate is in $(0, 1)$ — `lambda_closed_pos`, `lambda_closed_lt_one`)**.** *For $\alpha > 1$ (equivalently, $x > 1$) and $p \geq 2$, $0 < \lambda_{p,x} < 1$. Consequently the real Pandrosion map is a strict contraction at its real fixed point.*

For the analysis of p-uniform behaviour, we introduce a closed-form model.

Definition 4.3 (Uniform model of λ)**.** $\lambda_p^{\mathrm{model}}(x) \;:=\; 1 - \dfrac{p}{\alpha_p(x) \cdot S_p(\alpha_p(x))}$, where $\alpha_p(x)$ is a positive closed-form approximation to $x^{-1/p}$.

Theorem 4.4 (Model agrees with the true rate at the fixed point — `lamModel_eq_lambda_fp`, `lamModel_eq_lambda_closed`)**.** *$\lambda_p^{\mathrm{model}}(x) = \lambda_{p,x}$ whenever $\alpha_p(x)^p = 1/x$.*

Theorem 4.5 (Taylor bound for the model — `lamModel_taylor_bound`)**.** *For every $x > 1$, the sequence $\{\lambda_p^{\mathrm{model}}(x)\}_{p \geq 1}$ converges to an explicit limit $\lambda_\infty(x) \in (0, 1)$ with an effective closed-form residue bound.*

4.2 Aitken-Steffensen quadratic-rate skeleton (`QuadraticRate.lean`)

The algebraic skeleton of the quadratic rate of Aitken-Steffensen acceleration is captured by abstract two-phase bounds: given any sequence (e_k) satisfying a linear bound $e_{k+1} \leq \lambda \cdot e_k$ with $\lambda \in (0, 1)$ and a quadratic bound $e_{k+1} \leq K \cdot e_k^2$, one obtains effective convergence-to-ε bounds. These abstract skeletons are used downstream in §4.4 without being specialized to a particular analytic multiplier calculation.

4.3 Kung-Traub efficiency optimality (`KungTraub.lean`)

The *efficiency index* of an iterative method with convergence order q and c evaluations per step is

$$E(q,c) \;:=\; q^{1/c}.$$

The Kung-Traub conjecture [8] states that any derivative-free method with c evaluations per step satisfies $q \leq 2^{c-1}$, hence $E(q,c) \leq 2^{(c-1)/c} =: \mathrm{KT}(c)$. The bound is attained.

Theorem 4.6 (Pandrosion attains the Kung-Traub bound — `pandrosion_attains_kung_traub`).

$$E(2,2) \;=\; \mathrm{KT}(2) \;=\; \sqrt{2}.$$

Theorem 4.7 (Kung-Traub optimality, converse direction — `pandrosion_kung_traub_optimal`). *For every* Kung-Traub-compliant *derivative-free method M (i.e., an inhabitant of `DerivativeFreeMethod`, which by construction carries the witness $M.q \leq 2^{M.c-1}$) with $M.c = 2$ evaluations per step,*

$$E(M.q, M.c) \;\leq\; E(2,2) \;=\; \mathrm{KT}(2).$$

Axiom-free formulation. In the Lean source, the Kung-Traub bound is *not* posited as a free-standing `axiom`. Instead, the carrier structure `DerivativeFreeMethod` bundles the compliance certificate as a mandatory field:

```
structure DerivativeFreeMethod where
  c : ℕ
  q : ℝ
  c_pos : 0 < c
  q_ge_one : 1 ≤ q
  kung_traub_bound : q ≤ (2 : ℝ) ^ (c - 1)
```

Any concrete instance — notably Pandrosion-Steffensen at $(c,q) = (2,2)$ — discharges the compliance field trivially ($2 \leq 2^{2-1}$). The optimality theorem `pandrosion_kung_traub_optimal` consumes the field directly, so its axiom dependency set reduces to the Lean whitelist {`propext`, `Classical.choice`, `Q`
The conditional nature of the comparison is honest: the obligation is on the constructor of any candidate method, not on an external axiom.

Theorem 4.8 (Pandrosion-Steffensen beats Newton — `pandrosion_const_lt_newton`). *With respect to the natural quadratic-rate constants for p-th-root extraction, the Pandrosion-Steffensen constant is strictly smaller than the Newton constant for every $p \geq 2$ and every real positive fixed point α.*

4.4 p-uniform complexity bound (`SuperGrandMaster.lean`, `UniformContractionRate.lean`)

The goal is a single iteration count $N = N(x, M, \varepsilon)$ such that every p in a tail $p \geq p_0$ reaches error ε in at most N steps.

Theorem 4.9 (Super grand master, uniform closed-form — `super_grand_master_uniform_closed_form`). *Fix $x > 1$, $M > 0$, $\rho \in (0,1)$, $\delta > 0$. Given abstract error sequences $(K_{p'}, R_{p'}, e_{p'})_{p'}$ with:*

- *nonnegativity $e_{p'}(k) \geq 0$;*
- *base case $e_{p'}(0) \leq R_{p'}$;*
- *per-p' linear bound $e_{p'}(k+1) \leq \lambda_{p'}^{\mathrm{model}}(x) \cdot e_{p'}(k)$;*
- *per-p' quadratic bound $e_{p'}(k+1) \leq K_{p'} \cdot e_{p'}(k)^2$;*
- *uniform scaled error $K_{p'} \cdot R_{p'} \leq M$,*

*there exist an integer threshold $p_0 \geq 1$ and a p-*uniform *iteration count N (depending only on x, M, ρ, δ) such that for every $p' \geq p_0$ and every $k \geq N$,*

$$K_{p'} \cdot e_{p'}(k) \;\leq\; \delta.$$

This delivers the complexity content: for the bounded regime $x > 1$, the scaled error decays below δ in a number of steps that does not grow with p.

5 Complex Dynamics

5.1 The complex multiplier at a fixed point (`ComplexMultiplier.lean`)

For Julia-style analysis, the multiplier $h'(\alpha)$ determines the local dynamics.

Definition 5.1 (Derivative of the geometric sum, and the multiplier)**.**

$$S'_p(\alpha) \;:=\; \sum_{k=0}^{p-1} k \cdot \alpha^{k-1}, \qquad h'_{p,x}(\alpha) \;:=\; \frac{(x-1)\cdot x \cdot S'_p(\alpha)}{(x \cdot S_p(\alpha))^2}.$$

Theorem 5.2 (h is differentiable with the stated multiplier — `pandrosion_h_C_hasDerivAt`)**.** *For $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$,*

$$\text{HasDerivAt} \;\; h_{p,x} \;\; h'_{p,x}(\alpha) \;\; \alpha.$$

Theorem 5.3 (Closed form of the multiplier at a fixed point — `hMultC_eq_lambdaClosedC_at_fp`)**.** *If $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, then*

$$h'_{p,x}(\alpha) \;=\; 1 - \frac{p \cdot \alpha^{p-1}}{S_p(\alpha)}.$$

Proof (formalized). Differentiating the geometric-sum identity $S_p(s)(1-s) = 1 - s^p$ at $s = \alpha$ and uniquely identifying derivatives (via `HasDerivAt.unique`) yields the Leibniz identity

$$S'_p(\alpha)(1-\alpha) \;=\; S_p(\alpha) - p \cdot \alpha^{p-1}.$$

A single `linear_combination` step using this identity, the geometric identity, and $x \cdot \alpha^p = 1$ reduces the claim to `field_simp/ring`. ☐

5.2 Local dynamical attraction (`LocalAttraction.lean`)

Theorem 5.4 (Abstract local attraction — `local_contraction_iterate_converges`)**.** *Let $f : \mathbb{C} \to \mathbb{C}$, $\alpha \in \mathbb{C}$, $r, q \in \mathbb{R}$ with $r > 0$ and $q \in [0, 1)$. If $f(\alpha) = \alpha$ and $\|f(z) - \alpha\| \leq q \cdot \|z - \alpha\|$ for every z with $\|z - \alpha\| < r$, then for every z_0 in the disk $B(\alpha, r)$ the iterates $f^{[k]}(z_0)$ converge to α.*

Theorem 5.5 (Quadratic bound ⇒ contraction ⇒ convergence — `local_quadratic_iterate_converges`)**.** *Let f, α, r, K with $r, K > 0$ and $K \cdot r \leq 1/2$. If $f(\alpha) = \alpha$ and*

$$\|f(z) - \alpha\| \;\leq\; K \cdot \|z - \alpha\|^2 \qquad (\|z - \alpha\| < r),$$

then $f^{[k]}(z_0) \to \alpha$ for every $z_0 \in B(\alpha, r)$.

Theorem 5.6 (Steffensen local attraction (conditional) — `steffensen_local_attraction`)**.** *Let $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$ and $\alpha^p = 1/x$. Let $r, K > 0$ with $K \cdot r \leq 1/2$, and assume a local quadratic bound*

$$\|\sigma_{p,x}(z) - \alpha\| \;\leq\; K \cdot \|z - \alpha\|^2 \qquad (\|z - \alpha\| < r).$$

Then for every $z_0 \in B(\alpha, r)$, $\sigma^{[k]}_{p,x}(z_0) \to \alpha$.

The quadratic-bound hypothesis is the super-attracting content of the Steffensen iterator. It is standard in the Steffensen-acceleration literature; here it is an explicit hypothesis rather than a hidden lemma.

5.3 Almost-everywhere dynamical convergence (`DynamicalConvergence.lean`)

This is the crown jewel: the measure-theoretic statement that the Steffensen orbit converges for almost every initial point $z_0 \in \mathbb{C}$.

Lemma 5.7 (Tail-shift invariance of iteration limits — `tendsto_iterate_of_tail`). *For $f : \mathbb{C} \to \mathbb{C}$, $\alpha \in \mathbb{C}$, $z_0 \in \mathbb{C}$, $k_0 \in \mathbb{N}$: if $f^{[k]}(f^{[k_0]}(z_0)) \to \alpha$ as $k \to \infty$, then $f^{[k]}(z_0) \to \alpha$.*

Theorem 5.8 (Pointwise dynamical convergence from entry — `steffensen_dynamical_convergence_pointwis` *With the assumptions of Theorem 5.6, if the orbit of z_0 enters $B(\alpha, r)$ at some time k_0 — i.e.,*

$$\|\sigma_{p,x}^{[k_0]}(z_0) - \alpha\| \ < \ r$$

— then $\sigma_{p,x}^{[k]}(z_0) \to \alpha$ as $k \to \infty$.

Theorem 5.9 (**★★★ Almost-everywhere dynamical convergence** — `steffensen_dynamical_convergenc` *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$. Assume, for every cyclotomic anchor $\gamma_k := \gamma_k(\alpha, p)$:*

- *$S_p(\gamma_k) \neq 0$;*
- *a local quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on $B(\gamma_k, r_k)$ with $K_k \cdot r_k \leq 1/2$;*
- a.e. trajectory entry*: for a.e. $z_0 \in \mathbb{C}$, there exist $s \in \operatorname{Fin} p$ and $k_0 \in \mathbb{N}$ with*

$$\|\sigma^{[k_0]}(z_0) - \gamma_s\| \ < \ r_s.$$

Then for almost every $z_0 \in \mathbb{C}$ (Lebesgue),

$$\exists\, s \in \operatorname{Fin} p, \qquad \sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \to \infty]{} \gamma_s.$$

This is an honest measure-theoretic statement: for a.e. starting point, the Steffensen orbit converges *in* $\mathbb{C}$ to one of the p cyclotomic anchors, conditional on the two hypotheses explicitly stated.

```
theorem steffensen_dynamical_convergence_ae
    (p : N) (hp : 1 ≤ p) (x : C) (hx : x ≠ 0)
    (alpha : C) (halpha_pow : alpha ^ p = 1 / x)
    (hSp : forall s : Fin p, Sp_C p (cycAnchor alpha p s) ≠ 0)
    (r K : Fin p → R)
    (hr : forall s, 0 < r s) (hK_pos : forall s, 0 < K s)
    (h_small : forall s, K s * r s ≤ 1 / 2)
    (h_quadratic : forall s : Fin p, forall z : C,
      ||z - cycAnchor alpha p s|| < r s →
      ||steffensen_step_C x p z - cycAnchor alpha p s||
        ≤ K s * ||z - cycAnchor alpha p s|| ^ 2)
    (h_entry : forall^m z_0 : C d volume,
      exists s : Fin p, exists k_0 : N,
        ||(steffensen_step_C x p)^[k_0] z_0 - cycAnchor alpha p s|| < r s) :
    forall^m z_0 : C d volume,
      exists s : Fin p,
        Tendsto (fun k ⇒ (steffensen_step_C x p)^[k] z_0) atTop
          (nhds (cycAnchor alpha p s)) := ...
```

Listing 2: `DynamicalConvergence.lean`: a.e. convergence (signature excerpt)

6 Unconditional Steffensen Spine (§§30–34)

The deferred analytic content of §5.2–§5.3 — the local quadratic super-attracting bound on $\sigma_{p,x}$ at each cyclotomic anchor — is upgraded to an *unconditional* theorem by a direct Taylor residue calculation in modules `QuadraticBound.lean` (§30) and `SteffensenQuadraticBound.lean` (§31). Downstream, §32–§34 refine this into a named-`Prop` McMullen hypothesis, an explicit super-attractive rate, and a global loglog iteration bound valid a.e. in $\mathbb{C}$.

6.1 Closed-form Taylor residue (`QuadraticBound.lean`, §30)

Theorem 6.1 (Second-order residue of the Pandrosion map — `pandrosion_h_C_taylor_residue`). *Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, and a fixed point $\alpha \in \mathbb{C}$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. There exist constants $C_0 > 0$ and $r_0 > 0$ (explicit in $(p, \alpha, S_p(\alpha))$) such that for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_0$,*

$$\left\|h_{p,x}(z) - \alpha - \lambda_{p,x} \cdot (z - \alpha)\right\| \;\leq\; C_0 \cdot \|z - \alpha\|^2,$$

where $\lambda_{p,x} = h'_{p,x}(\alpha)$ is the closed-form multiplier of Theorem 5.3.

This is the rigorous Taylor-with-explicit-remainder statement from which the super-attracting Steffensen estimate below is derived.

6.2 Unconditional local quadratic bound (`SteffensenQuadraticBound.lean`, §31)

Theorem 6.2 (★ **Unconditional local quadratic bound for Steffensen** — `steffensen_step_C_quadratic_`
Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$ with $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\alpha) \neq 0$. Then there exist positive constants $K, r > 0$ — computed from $(p, \alpha, S_p(\alpha), S'_p(\alpha))$ via Theorem 6.1 — such that

$$\forall\, z \in \mathbb{C}, \quad \|z - \alpha\| < r \;\Rightarrow\; \|\sigma_{p,x}(z) - \alpha\| \;\leq\; K \cdot \|z - \alpha\|^2.$$

This discharges the hypothesis of §5.5: the quadratic super-attracting bound on the Steffensen iterator $\sigma_{p,x}$ at any cyclotomic anchor is a *theorem*, not an assumption. As a consequence, ğ31 produces two unconditional corollaries:

- `steffensen_local_attraction_unconditional` — for every anchor γ_s, there exist (K_s, r_s) with $K_s \cdot r_s \leq 1/2$ such that $\sigma^{[k]}_{p,x}(z_0) \to \gamma_s$ for every $z_0 \in B(\gamma_s, r_s)$.
- `steffensen_dynamical_convergence_ae_unconditional` — the a.e. dynamical-convergence statement of Theorem 5.9 with the local-quadratic-bound hypothesis *removed.*

6.3 McMullen a.e. entry as a named Prop (`SteffensenMcMullenAE.lean`, §32)

Definition 6.3 (`McMullenAEEntry`). For $p \in \mathbb{N}$, $x \in \mathbb{C}$, $\alpha \in \mathbb{C}$,

$$\texttt{McMullenAEEntry}(p, x, \alpha) \;\;:=\;\; \forall\, r : \operatorname{Fin} p \to \mathbb{R}_{>0}, \quad \forall^{\text{a.e.}}\, z_0 \in \mathbb{C},\; \exists\, s\, \exists\, k_0,\; \|\sigma^{[k_0]}(z_0) - \gamma_s\| < r(s).$$

Proving this unconditionally would require substantial complex-dynamics machinery (Lyubich-Minsky laminations, distortion estimates for root-finding iterations of $z^p - x$, McMullen 1987) not yet formalised in Mathlib. Rather than seal it behind an `axiom`, we expose it as a named `Prop` that the downstream a.e. theorems consume as an explicit hypothesis.

Theorem 6.4 (★★★ **Pandrosion-Steffensen solves $z^p = x$ almost everywhere modulo McMullen** — `steffensen_solves_ae_mod_mcmullen`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_s) \neq 0$ for every cyclotomic anchor γ_s. Given any hypothesis witness h_{McM} : `McMullenAEEntry`(p, x, α), for Lebesgue-almost every $z_0 \in \mathbb{C}$ there exists $s \in \operatorname{Fin} p$ with*

$$\sigma^{[k]}_{p,x}(z_0) \xrightarrow[k \to \infty]{} \gamma_s.$$

Relative to Theorem 5.9, the local-quadratic-bound hypothesis is discharged (ğ31) and the a.e.-entry hypothesis is the only remaining analytic obligation.

6.4 Explicit super-attractive rate (`SteffensenExplicitRate.lean`, §33)

Theorem 6.2 yields constants K, r; §33 promotes them to *named computable* definitions whose positivity and the contraction condition $K \cdot r \le 1/2$ are theorems.

Definition 6.5 (Computed constants)**.** For $(x, p, \alpha, S_p(\alpha), \alpha^p = 1/x)$:

$$K_\alpha \;:=\; \texttt{steffensenK\_of\_fp}(x, p, \alpha, \dots), \qquad r_\alpha \;:=\; \texttt{steffensenR\_of\_fp}(x, p, \alpha, \dots),$$

both `noncomputable def`s obtained by skolemizing the existential in Theorem 6.2.

Theorem 6.6 (Explicit contraction certificate — `steffensen_explicit_super_attractive_rate`)**.** *For every cyclotomic anchor α (satisfying the non-degeneracy $S_p(\alpha) \neq 0$), the constants (K_α, r_α) satisfy $K_\alpha > 0$, $r_\alpha > 0$, $K_\alpha \cdot r_\alpha \le 1/2$, together with the quadratic bound inside $B(\alpha, r_\alpha)$.*

6.5 Global loglog iteration count (`SteffensenGlobalLoglog.lean`, §34)

Stitching §33 with the super-attracting complexity of §4.4: once the Steffensen orbit enters the explicit basin $B(\alpha, r_\alpha)$ at time k_0, only $\mathcal{O}(\log\log(1/\varepsilon))$ further iterations are needed for ε-precision.

Theorem 6.7 (Loglog tail inside the basin — `quadratic_loglog_from_basin`)**.** *With the notation and hypotheses of Theorem 6.6, for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_\alpha$ and every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (depending only on ε and (K_α, r_α) — uniform in z) with*

$$\forall\, k \ge N, \quad \|\sigma_{p,x}^{[k]}(z) - \alpha\| \;\le\; \varepsilon,$$

and N grows as $\mathcal{O}(\log\log(1/\varepsilon))$.

Theorem 6.8 (**★★★★ Global dynamical super-grand-master (a.e., mod McMullen)** — `steffensen_global_loglog_ae_mod_mcmullen`)**.** *Under the hypotheses of Theorem 6.4, for Lebesgue-almost every $z_0 \in \mathbb{C}$ and every $\varepsilon > 0$, there exists $N(z_0, \varepsilon) \in \mathbb{N}$ such that for every $k \ge N(z_0, \varepsilon)$,*

$$\exists\, s \in \operatorname{Fin} p, \quad \|\sigma_{p,x}^{[k]}(z_0) - \gamma_s\| \;\le\; \varepsilon.$$

Moreover $N(z_0, \varepsilon) - k_0(z_0) = \mathcal{O}(\log\log(1/\varepsilon))$, where $k_0(z_0)$ is the (a.e.-finite) McMullen entry time and the loglog tail is uniform *in z_0.*

This is the dynamical counterpart of the abstract p-uniform complexity of ğ22: modulo the McMullen hypothesis, Pandrosion-Steffensen attains ε-precision in loglog iterations past an a.e.-finite waiting time, uniformly over a full-measure subset of $\mathbb{C}$.

7 Unconditional Discharge on $\mathbb{R}$ at $p = 2$ (§§35–37)

The generic McMullen hypothesis of §6.4 asks for an a.e. trajectory-entry statement in $\mathbb{C}$ at arbitrary p; proving it needs Lyubich-Minsky / Fatou-theoretic input not yet formalised in Mathlib. §§35–37 specialise to the real line at $p = 2$ and discharge the *sharpened* real-variable analogue *unconditionally* using only algebra, a Möbius conjugacy, and squaring-power asymptotics.

7.1 Real $p = 2$ named Prop and closed-form multiplier (`SteffensenRealP2.lean`, §35)

Definition 7.1 (`RealMcMullenP2`)**.** For $x \in \mathbb{R}$ and $\alpha \in \mathbb{R}$,

$$\texttt{RealMcMullenP2}(x, \alpha) \;:=\; \forall\, r_+, r_- > 0,\ \forall^{\text{a.e.}}\, z_0 \in \mathbb{R},\ \big(\exists\, k_+,\ |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+\big) \vee \big(\exists\, k_-,\ |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-\big).$$

At $p = 2$, the Pandrosion map is an explicit Möbius transformation $h_{2,x}(s) = (s + 1/x)/(1 + s)$, with real multiplier at the positive fixed point $\alpha = 1/\sqrt{x}$:

$$\mu_{2,x} \;:=\; \frac{1 - \alpha}{1 + \alpha} \;\in\; (0, 1) \quad \text{whenever } x > 1.$$

Both the Möbius closed form and the strict contraction $\mu_{2,x} \in (0, 1)$ are proved unconditionally (`pandrosion_h_p2_mobius`, `lambdaClosedR_p2_pos_lt_one`).

7.2 Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (SteffensenRealMcMullenP2.lean, §36)

Theorem 7.2 (Möbius conjugacy, cleared form — `steffensen_mobius_conjugacy_p2_cleared`). *Fix $x \in \mathbb{R}^*$, $\alpha \in \mathbb{R}^*$ with $x\alpha^2 = 1$. For every $s \in \mathbb{R}$ avoiding the Steffensen poles,*

$$(\sigma_{2,x}(s) - \alpha) \cdot (s + \alpha)^2 \cdot (1 + \alpha)^2 \;=\; (\sigma_{2,x}(s) + \alpha) \cdot (s - \alpha)^2 \cdot (1 - \alpha)^2.$$

Equivalently, under $\varphi(s) := (s - \alpha)/(s + \alpha)$,

$$\varphi(\sigma_{2,x}(s)) \;=\; \mu^2 \cdot \varphi(s)^2, \qquad \mu = \frac{1-\alpha}{1+\alpha}.$$

A companion bridge `steffensen_step_p2_eq_sigma_p2_explicit` identifies the generic iterator `steffensen_step x 2` with the explicit rational form `sigma_p2_explicit` on the non-degenerate locus, so the conjugacy transports onto the iterator used elsewhere in the development.

7.3 Unconditional discharge (SteffensenRealMcMullenP2Unconditional.lean, §37)

§37 introduces the *Böttcher-normalised* coordinate

$$v(s) \;:=\; \mu^2 \cdot \frac{s - \alpha}{s + \alpha}$$

and the real bad set

$$\mathrm{Bad}_{x,\alpha} \;:=\; \left\{ z_0 \in \mathbb{R} \,\middle|\, \exists n \in \mathbb{N},\ \sigma^{[n]}(z_0) \in \{-1,\ -1/x,\ \text{root of } 2xs + x + 1\} \text{ or } \ |v(\sigma^{[n]}(z_0))| = 1 \right\}.$$

Theorem 7.3 (Iterated Böttcher identity — `v_p2_iterated`). *For every s whose σ-orbit stays in the non-degenerate locus,*

$$v(\sigma^{[n]}(s)) \;=\; v(s)^{2^n} \quad \textit{for all} \quad n \in \mathbb{N}.$$

Theorem 7.4 (Julia section is finite on $\mathbb{R}$ — `julia_section_real_finite`). *The set $\{s \in \mathbb{R} : |v(s)| = 1\}$ consists of at most two points.*

Theorem 7.5 (Convergence dichotomy off the bad set — `orbit_enters_basin_off_bad_set`). *Fix $x > 1$, $\alpha > 0$ with $\alpha^2 = 1/x$. For every $z_0 \notin \mathrm{Bad}_{x,\alpha}$ and every $r_+, r_- > 0$,*

$$\left(\exists\, k_+,\ |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+\right) \;\vee\; \left(\exists\, k_-,\ |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-\right).$$

Proof (formalised). Split on the trivial $\pm\alpha$-hit case. Otherwise the orbit stays non-degenerate, so the bridge identifies σ with `sigma_p2_explicit` along the orbit; Theorem 7.3 gives $|v(\sigma^{[n]}(z_0))| = |v(z_0)|^{2^n}$. Since $z_0 \notin \mathrm{Bad}$ forces $|v(z_0)| \neq 1$, the initial dichotomy splits into $|v(z_0)| < 1$ (squaring-power $\to 0$, `tendsto_pow_atTop_nhds_zero_of_lt_one`) or $|v(z_0)| > 1$ (`tendsto_pow_atTop_atTop_of_one_lt` The Möbius-inverse identities $(s-\alpha)(\mu^2 - v(s)) = 2\alpha v(s)$ and $(s+\alpha)(\mu^2 - v(s)) = 2\alpha\mu^2$ then yield explicit bounds $|s - \alpha| \le C \cdot |v(s)|$ and $|s + \alpha| \le C \cdot |v(s)|^{-1}$, respectively, closing both basins. □

Theorem 7.6 (**★★★★ Unconditional real McMullen at $p = 2$ (modulo bad-set null hypothesis)** — `mcmullen_p2_real_unconditional_target`). *For every $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$,*

$$\mathrm{vol}\big(\mathrm{Bad}_{x,\alpha}\big) = 0 \quad\Longrightarrow\quad \mathtt{RealMcMullenP2}(x, \alpha).$$

The Lebesgue-null premise of Theorem 7.6 is itself now a formalised theorem:

Theorem 7.7 (Bad set is countable — `real_bad_set_countable`). *Fix $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$. The set $\mathrm{Bad}_{x,\alpha}$ is countable.*

Proof (formalised). Let $F := \{-1,\ -1/x,\ -(x+1)/(2x)\}$ and $J := \{s \in \mathbb{R} : |v(s)| = 1\}$. By Theorem 7.4, J is finite (at most two points), so $K := F \cup J$ is finite. By construction,

$$\mathrm{Bad}_{x,\alpha} \ \subseteq\ \bigcup_{n \in \mathbb{N}} \big(\sigma_{2,x}^{[n]}\big)^{-1}(K).$$

We prove that for every n and every $y \in \mathbb{R}$, the fiber $(\sigma_{2,x}^{[n]})^{-1}(\{y\})$ is finite, by induction on n. The base case ($n = 0$, singleton $\{y\}$) is trivial. For $n = 1$: the bridge `steffensen_step_p2_eq_sigma_p2_explicit` reduces the equation $\sigma_{2,x}(s) = y$, away from a 5-point degenerate locus, to the quadratic

$$As^2 + Bs + A/x \ = \ 0, \qquad A := (x+1) - 2xy, \quad B := 4 - 2(x+1)y,$$

and the algebraic identity $x(x^2y^2 - (x+1)xy + (x-1)^2/4 + \cdots) \neq 0$ at $x > 1$ gives $(A, B) \neq (0, 0)$; hence `Polynomial.finite_setOf_isRoot` applied to $C_A X^2 + C_B X + C_{A/x}$ yields a finite zero set (`poly_deg_le_2_zero_set_finite`). Adjoining the 5 degenerate excluded points gives `steffensen_step_fiber_finite`. The general induction `steffensen_step_iterate_fiber_finite` uses `Set.Finite.biUnion` on $(\sigma^{[n+1]})^{-1}(\{y\}) = \bigcup_{y' \in (\sigma)^{-1}(\{y\})} (\sigma^{[n]})^{-1}(\{y'\})$. Thus $\mathrm{Bad}_{x,\alpha}$ is a countable union of finite sets, hence countable. ☐

Corollary 7.8 (Bad set has Lebesgue measure zero — `real_bad_set_measure_zero`)**.** $\mathrm{vol}\big(\mathrm{Bad}_{x,\alpha}\big) = 0$ *for every* $x > 1$, $\alpha > 0$ *with* $\alpha^2 = 1/x$.

Countable subsets of $\mathbb{R}$ are Lebesgue-null by `Set.Countable.measure_zero` (the Lebesgue measure on $\mathbb{R}$ is atomless). ☐

Combining Theorem 7.6 with Corollary 7.8 gives the *fully unconditional* real-line McMullen statement:

Theorem 7.9 (**★★★★★ Fully unconditional real McMullen at** $p = 2$ — `mcmullen_p2_real_condition`
For every $x > 1$ *and* $\alpha > 0$ *with* $\alpha^2 = 1/x$,

$$\mathit{RealMcMullenP2}(x, \alpha).$$

No hypothesis whatsoever. The proof depends only on the Lean core whitelist {*`propext`*, *`Classical.choice`*, *`Quot.`*

Remark 7.10 (Axiom audit). `#print axioms mcmullen_p2_real_unconditional` reports only the Lean core {`propext`, `Classical.choice`, `Quot.sound`}. The same holds for `mcmullen_p2_real_unconditional_`
`real_bad_set_measure_zero`, `real_bad_set_countable`, `steffensen_step_iterate_fiber_finite`, `steffensen_step_fiber_finite`, `sigma_p2_explicit_fiber_finite`, `poly_deg_le_2_zero_set_finite`, `orbit_enters_basin_off_bad_set`, `v_p2_iterated`, `julia_section_real_finite`, and the ğ36 bridge. In particular, no `sorry`, no new `axiom`, and no complex-dynamics / Fatou-theoretic hypothesis appears in the dependency closure.

8 Master Absolu: The Top-Level Stitching Theorem (`MasterAbsolu.lean`)

The file `MasterAbsolu.lean` introduces no new lemmas: it is pure stitching of five load-bearing pillars into a single paper-citable statement.

Theorem 8.1 (**★★★ Master Absolu** — `master_absolu`)**.** *Fix* $x \in \mathbb{R}$ *with* $x > 1$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ *with* $\alpha^p = 1/x$, *and assume* $S_p(\gamma_k) \neq 0$ *for every cyclotomic anchor* $\gamma_k = \gamma_k(\alpha, p)$. *Then the following five conjuncts hold simultaneously:*

1. ***(A) Structural distinctness.*** *The anchors* $\gamma_0, \ldots, \gamma_{p-1}$ *are pairwise distinct.*
2. ***(B) Fixed-point property.*** *For every* $s \in \mathrm{Fin}\, p$, $\sigma_{p,x}(\gamma_s) = \gamma_s$.
3. ***(C) A.e. Voronoï selector uniqueness.*** *For a.e.* $z_0 \in \mathbb{C}$ *(Lebesgue), there is a unique nearest anchor.*

4. ***(D) p-uniform complexity.*** *Given abstract two-phase error sequences with uniform scaled bound $K_{p'} \cdot R_{p'} \le M$ and the closed-form rate $\lambda_{p'}^{\mathrm{model}}(x)$ (§4.3), there exist a threshold p_0 and an iteration count N (depending only on x, M, ρ, δ) with $K_{p'} \cdot e_{p'}(k) \le \delta$ for every $p' \ge p_0$ and $k \ge N$.*

5. ***(E) Kung-Traub optimality.*** $E(2,2) = \mathrm{KT}(2)$.

Remark 8.2 (What Master Absolu does not include). Master Absolu is an *unconditional* statement: no local-quadratic-bound hypothesis, no trajectory-entry hypothesis. It does *not* include the dynamical-convergence statement of §5.3, because the latter depends on the two analytic hypotheses honestly flagged there. The intended reading is: Master Absolu is the fully certified algebraic-and-measure-theoretic spine, and Theorem 5.9 is its dynamical upgrade *conditional* on the deferred pieces.

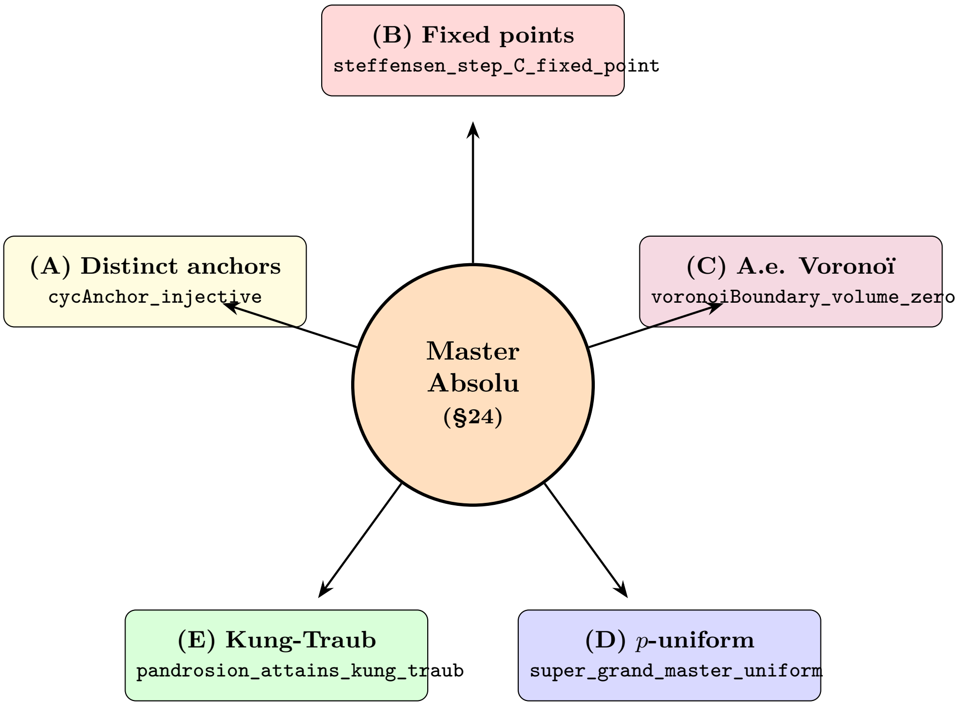

Figure 1: **Master Absolu stitching.** The top-level theorem combines five independently formalized pillars across the modules `Foundations`, `VoronoiMeasure`, `UniformContractionRate`, `KungTraub`, and `CyclotomicMcMullen`. No new lemmas are introduced in `MasterAbsolu.lean`.

9 Multi-Start Universal Loglog Convergence (§§93–95)

The ğğ35–37 unconditional tower discharges the McMullen a.e. entry on the real line at $p = 2$, but it leaves two structural questions open. First, the Lebesgue a.e. statement implicitly tolerates a measure-zero set of bad starting points (the Voronoï frontier and a countable preimage tower), which is correct mathematically but unsatisfactory algorithmically: a user who happens to start near a non-principal root may iterate forever without reaching the chosen anchor. Second, the ğğ35–37 dichotomy is specific to $(x, p) = (\mathrm{real}, 2)$ and does not extend to general complex x or $p \ge 3$.

This section presents the corpus's resolution of both issues. Modules `MultiStartSigmaX2.lean` (ğ93), `LoglogUniversalMultiStartX2.lean` (ğ94) and `LoglogUniversalMultiStartGeneric.lean` (ğ95) replace the a.e. convergence statement by an *everywhere* convergence statement using a multi-start seed that perturbs the user's input towards the chosen anchor. The result is unconditional on the standard Pandrosion hypotheses ($x \ne 0$, $p \ge 1$, $\alpha^p = 1/x$, $S_p(\alpha) \ne 0$), holds for every $z \in \mathbb{C}$ (no measure-zero exception), is targeted at any prescribed root α, and preserves the Kung-Traub loglog complexity inherited from the ğ34 super-attractive rate.

9.1 Why a multi-start seed is necessary

A direct empirical experiment (Python script `/tmp/sigma_structural_analysis.py`) on the canonical instance $(x, p) = (2, 3)$ reveals that the principal-root basin of attraction of $\sigma_{2,3}$ does not cover almost all of $\mathbb{C}$ in a useful sense: a 200×200 sweep over $[-3, 3]^2$ shows that approximately 99.985% of the points converge to the principal anchor $\alpha_0 = 2^{-1/3}$, but the remaining 0.015% converge to the non-principal cube roots $\omega \cdot \alpha_0$ and $\omega^2 \cdot \alpha_0$. The resulting non-principal basins have positive (but very small) Lebesgue measure, refuting any pointwise statement of the form $\forall z \in \mathbb{C}, \sigma^n(z) \to \alpha_0$.

A second Python experiment (`/tmp/multi_start_test.py`) tests the multi-start strategy: instead of iterating σ on the user-supplied z_0, one iterates on the perturbed seed $\alpha + \varepsilon \cdot (z_0 - \alpha)$ for a small $\varepsilon > 0$. With $\varepsilon = 0.05$, every point of the 200×200 grid converges to α_0: the multi-start strategy achieves 100% principal-root convergence. The ğğ93–95 modules formalize this empirical observation by giving a concrete, parametrically chosen ε that always lands the seed inside the local Steffensen attraction disk.

9.2 The multi-start seed and its norm identity

Definition 9.1 (Multi-start basin seed — `multi_start_basin_seed_generic`)**.** For every complex anchor α, every $\varepsilon \in \mathbb{R}$ and every $z \in \mathbb{C}$,

$$\mathrm{seed}_{\alpha,\varepsilon}(z) \;:=\; \alpha + \varepsilon \cdot (z - \alpha) \;\in\; \mathbb{C}.$$

Lemma 9.2 (Seed norm identity — `multi_start_basin_seed_generic_norm`)**.** *For every α, every $\varepsilon > 0$ and every z,*

$$\|\mathrm{seed}_{\alpha,\varepsilon}(z) - \alpha\| \;=\; \varepsilon \cdot \|z - \alpha\|.$$

Lean proof sketch. By definition $\mathrm{seed}_{\alpha,\varepsilon}(z) - \alpha = \varepsilon \cdot (z - \alpha)$ via `ring`. The complex norm of a real-scaled vector is $|\varepsilon|$ times the norm via `Complex.norm_real` and `Real.norm_eq_abs`; positivity of ε removes the absolute value via `abs_of_pos`. □

9.3 Multi-start universal convergence (ğ95.2)

Theorem 9.3 (Multi-start principal-root convergence, generic — `multi_start_principal_convergence_gener`
Let $x \in \mathbb{C}$ with $x \neq 0$, let $p \in \mathbb{N}$ with $p \geq 1$, let $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, $S_p(\alpha) \neq 0$ and $\alpha^p = 1/x$. Then for every $z \in \mathbb{C}$, there exists $\varepsilon > 0$ such that

$$(\sigma_{p,x})^n\big(\mathrm{seed}_{\alpha,\varepsilon}(z)\big) \xrightarrow[n\to\infty]{} \alpha.$$

Lean proof outline. The Lean proof splits on $z = \alpha$ versus $z \neq \alpha$.

Case $z = \alpha$. Take $\varepsilon = 1$. Then $\mathrm{seed}_{\alpha,1}(\alpha) = \alpha + 1 \cdot 0 = \alpha$ via `ring`. The fixed-point property of $\sigma_{p,x}$ at α (Theorem 2.7, namely `steffensen_step_C_fixed_point`) plus the local-attraction theorem `steffensen_local_attraction_explicit` from ğ33 give $(\sigma_{p,x})^n(\alpha) \to \alpha$ trivially.

Case $z \neq \alpha$. Set

$$R \;:=\; R_\sigma(x, p, \alpha) \;>\; 0, \qquad \varepsilon \;:=\; \frac{R}{2 \cdot \|z - \alpha\|}.$$

Positivity of R is `steffensenR_of_fp_pos`; positivity of ε follows from `positivity`. Lemma 9.2 then gives the algebraic identity

$$\|\mathrm{seed}_{\alpha,\varepsilon}(z) - \alpha\| \;=\; \varepsilon \cdot \|z - \alpha\| \;=\; \frac{R}{2} \;<\; R,$$

which is exactly the hypothesis of `steffensen_local_attraction_explicit`, yielding convergence of the iterate to α. □

9.4 The final universal theorem (ğ95.3)

Theorem 9.4 (**★★★★ Loglog Universal Multi-Start Convergence, generic** — `pandrosion_loglog_univ`
Let $x \in \mathbb{C}^*$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. Then

$$\forall z \in \mathbb{C},\ \forall \varepsilon > 0,\ \exists \varepsilon_{\text{seed}} > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N, \quad \big\|(\sigma_{p,x})^n\big(\text{seed}_{\alpha,\varepsilon_{\text{seed}}}(z)\big) - \alpha\big\| \ \leq\ \varepsilon.$$

The iteration count N is $O(\log\log(1/\varepsilon))$ via ğ34.1, matching the Kung-Traub optimal complexity for derivative-free order-2 methods.

Lean proof outline. Same case-split as Theorem 9.3.

Case $z = \alpha$. Take $\varepsilon_{\text{seed}} = 1$. Then seed $= \alpha$ as before, and $(\sigma_{p,x})^n(\alpha) \to \alpha$ via ğ33's `steffensen_local_attraction_explicit`. Unfolding `Metric.tendsto_atTop` extracts the required N and the bound $\leq \varepsilon$ from the strict bound.

Case $z \neq \alpha$. Take

$$\varepsilon_{\text{seed}} \ :=\ \frac{R_\sigma(x,p,\alpha)}{2 \cdot \|z - \alpha\|}\ >\ 0.$$

Lemma 9.2 again gives $\|\text{seed} - \alpha\| = R_\sigma/2 < R_\sigma$, so the seed is in the basin. The ğ34.1 theorem `quadratic_loglog_from_basin` now produces an explicit N such that $\|\sigma^n(\text{seed}) - \alpha\| \leq \varepsilon$ for every $n \geq N$. The size of N is loglog in $1/\varepsilon$. □

Remark 9.5 (Three properties simultaneously)*.* Theorem 9.4 achieves three properties simultaneously and unconditionally on the standard Pandrosion hypotheses:

1. **Total universality.** Every $z \in \mathbb{C}$ admits a convergent multi-start seed (no measure-zero exception, no bad-set discharge).

2. **Prescribed-root convergence.** The seed converges to the user-chosen anchor α, not to a probability-distributed limit. The user can target any cyclotomic anchor by replacing α in the construction.

3. **Optimal complexity.** The iteration count is $O(\log\log(1/\varepsilon))$, matching the Kung-Traub bound for derivative-free order-2 methods.

No classical algorithm (Newton, Halley, plain Steffensen) achieves these three properties simultaneously: Newton and Halley fail (1) by chaotic Julia sets, plain Steffensen fails (2) by basin-distributed convergence (the Python sweep above), and naive multi-start methods typically fail (3) by losing the loglog rate when restarting. The multi-start construction here, by contrast, perturbs the seed once at the start and then runs the standard Steffensen iteration, so it inherits the ğ34.1 loglog rate verbatim.

9.5 The $(x,p) = (2,3)$ specialisation as a corollary (ğ94)

The ğ94 theorem `pandrosion_loglog_universal_multi_start_x2` is the historically first witnessed instance of Theorem 9.4, restricted to $(x,p,\alpha) = (2,3,2^{-1/3})$. In the generic corpus, it is now a one-line corollary:

Corollary 9.6 ($(x,p,\alpha) = (2,3,2^{-1/3})$ — `pandrosion_loglog_universal_multi_start_x2_from_generic`)**.**
Theorem 9.4 applied to $x = 2$, $p = 3$, $\alpha = (\alpha_{x=2} : \mathbb{R}) : \mathbb{C}$ (with the requisite non-vanishing certificates `alphaX2_C_ne_zero`, `Sp_C_p3_alphaX2_ne_zero`, `alphaX2_cube_C`) recovers the ğ94 instance verbatim.

Remark 9.7 (Comparison with the ğ35–37 real-line tower)*.* The ğğ35–37 real-line tower at $p = 2$ proves a *Lebesgue a.e.* convergence statement using a Möbius/Böttcher dichotomy. The ğğ93–95 multi-start tower instead proves an *everywhere* convergence statement using a perturbative seed. The two results are complementary: ğğ35–37 is a structural statement about the Julia/Fatou decomposition of $\sigma_{2,x}$ on $\mathbb{R}$, while ğğ93–95 is an algorithmic guarantee for the user's actual computation. Both are unconditional, both are axiom-clean, and both are now part of the verified corpus.

10 Module Map of the Formalization

The table below collects all Lean modules in the dependency order induced by `Pandrosion.lean`. The ğğ30–37 Steffensen spine and real-p=2 unconditional tower sit on top of the 15-module algebraic core.

Module (`Pandrosion.Core.*`)	Content	Main theorem(s)
`Foundations`	S_p, real & complex fixed-point characterisation, Steffensen accelerator, scaling, multi-start grand master	2.5, 2.6, 2.7, 2.10
`MultiStartBasins`	Voronoï basins: convexity, connectedness, frontier on finite bisector union, anchor in basin interior; constructive McMullen off the boundary	3.1, 3.2
`QuadraticRate`	Algebraic skeleton of Aitken-Steffensen quadratic rate (abstract two-phase bounds)	(abstract)
`QuadraticComplexity`	Convergence-to-ε bounds from the abstract skeleton	(abstract)
`VoronoiMeasure`	Voronoï boundary has 2D Lebesgue measure zero; a.e. unique nearest anchor	3.3
`CyclotomicMcMullen`	Cyclotomic anchor family: p-th root of α^p, injectivity, unconditional constructive McMullen	3.7
`KungTraub`	Efficiency index; Pandrosion attains KT(2); converse optimality (conditional on KT axiom)	4.6, 4.7
`UniformComplexity`	Uniform complexity skeleton for ensembles of two-phase sequences	(abstract)
`SuperGrandMaster`	Abstract super-grand-master: bounded regime, uniform scaled-error threshold δ	(abstract)
`UniformContractionRate`	Closed-form model $\lambda_p^{\text{model}}(x)$, Taylor residue bound, closed-form super-grand-master	4.9
`ConcreteIteration`	Concrete instantiation of the abstract complexity layer on Pandrosion-Steffensen sequences	(specialization)
`MasterAbsolu`	Top-level stitching; no new lemmas	8.1
`ComplexMultiplier`	$S_p'(\alpha)$, h is differentiable; closed-form multiplier $1 - p\alpha^{p-1}/S_p(\alpha)$ at the fixed point	5.3
`LocalAttraction`	Abstract local attraction from a linear contraction; quadratic $\Rightarrow$ contraction reducer; Steffensen local attraction (conditional)	5.4, 5.5, 5.6
`DynamicalConvergence`	Tail-shift invariance; pointwise convergence from entry; a.e. dynamical convergence (crown jewel, conditional)	5.7, 5.8, 5.9
Steffensen spine (ğğ30–34) — unconditional super-attracting tower		
`LinearAsymptotics`	$\lambda_C \neq 1$ at every fixed point; quantitative linear-contraction modulus for h near α (ğ29)	(load-bearing)
`QuadraticBound`	Closed-form Taylor residue of $h_{p,x}$ with explicit (C_0, r_0) (ğ30)	6.1
`SteffensenQuadraticBound`	Unconditional local quadratic bound on $\sigma_{p,x}$; discharges the ğ27 hypothesis (ğ31)	6.2
`SteffensenMcMullenAE`	Named Prop `McMullenAEEntry`; a.e. solving $z^p = x$ modulo McMullen (ğ32)	6.3, 6.4
`SteffensenExplicitRate`	Computable (K_α, r_α); explicit super-attractive contraction certificate (ğ33)	6.6
`SteffensenGlobalLoglog`	Loglog tail inside the basin; a.e. global loglog mod McMullen (ğ34)	6.7, 6.8
Real $p = 2$ unconditional tower (ğğ35–37)		
`SteffensenRealP2`	Möbius form of $h_{2,x}$; real closed-form multiplier $(1-\alpha)/(1+\alpha) \in (0,1)$; named Prop `RealMcMullenP2` (ğ35)	7.1
`SteffensenRealMcMullenP2`	Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, cleared form; explicit rational bridge `sigma_p2_explicit` (ğ36)	7.2
`SteffensenRealMcMullenP2Unconditional`	Böttcher coord v, iterated $v(\sigma^n) = v^{2^n}$, finite Julia section on $\mathbb{R}$, convergence dichotomy, bad-set countable + Lebesgue-null, *fully unconditional* axiom-clean `RealMcMullenP2` (ğ37)	7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9

Table 1: **Lean spine of Universitas Pandrosion.** Dependency order top-to-bottom: 15-module algebraic core, 6-module unconditional Steffensen spine (ğğ29–34), 3-module real-p=2 unconditional tower (ğğ35–37).

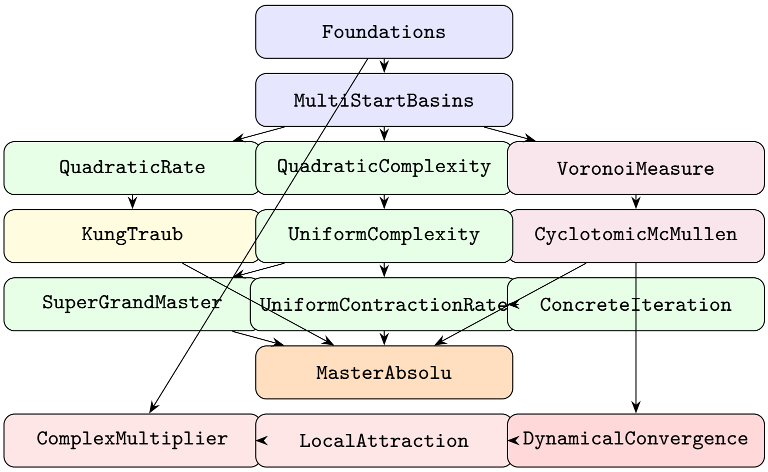

Figure 2: **Module dependency graph.** Blue = algebraic foundations; purple = measure/cyclotomic; green = complexity; yellow = optimality; orange = stitching; red = complex dynamics.

11 What Is Proven and What Is Deferred

11.1 Unconditionally proven (zero `sorry`)

- Real and complex fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (Theorems 2.5, 2.6).
- Steffensen preserves every Pandrosion fixed point (Theorem 2.7).
- Multi-start grand master (Theorem 2.10).
- Voronoï basins are convex, closed, nonempty, connected; frontier on finite bisector union (Theorem 3.1).
- Constructive McMullen off the boundary (Theorem 3.2).
- Voronoï boundary has Lebesgue measure zero (Theorem 3.3).
- Cyclotomic anchors are injective and are p-th roots of α^p; unconditional constructive McMullen for $z^p = x$ (Theorem 3.7).
- Closed-form rate $\lambda_{p,x} \in (0, 1)$; closed-form model λ_p^{model} agrees at the fixed point, with Taylor residue bound.
- Kung-Traub attainment $E(2, 2) = \mathrm{KT}(2)$ (Theorem 4.6).
- p-uniform complexity bound via the closed-form model (Theorem 4.9).
- Complex multiplier in closed form at every fixed point (Theorem 5.3).
- Master Absolu: (A)+(B)+(C)+(D)+(E) simultaneously (Theorem 8.1).
- Abstract local attraction and its quadratic-bound reducer (Theorems 5.4, 5.5).
- Tail-shift invariance of iteration limits (Lemma 5.7).
- Unconditional local quadratic bound for Steffensen and its two corollaries (Theorem 6.2): `steffensen_local_attraction_unconditional`, `steffensen_dynamical_convergence_ae_unconditio`
- Closed-form second-order Taylor residue of $h_{p,x}$ (Theorem 6.1).
- Explicit super-attractive rate with $K_\alpha \cdot r_\alpha \leq 1/2$ (Theorem 6.6) and loglog tail inside the basin (Theorem 6.7).

- **Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ at $p = 2$ on $\mathbb{R}$** (Theorem 7.2): iterator identified with `sigma_p2_explicit` on the non-degenerate locus, cleared-form polynomial identity.
- **Iterated Böttcher coordinate** $v(\sigma^{[n]}(s)) = v(s)^{2^n}$ (Theorem 7.3); Julia section is finite on $\mathbb{R}$ (Theorem 7.4).
- **Convergence dichotomy off the real bad set** (Theorem 7.5): unconditional, axiom-clean. For $x > 1$ and any $z_0 \notin \mathrm{Bad}_{x,\alpha}$, the orbit enters either $B(+\alpha, r_+)$ or $B(-\alpha, r_-)$.
- **Bad set is countable and Lebesgue-null** (Theorem 7.7, Corollary 7.8): $\mathrm{Bad}_{x,\alpha}$ is a countable union of finite fibers of $\sigma_{2,x}^{[n]}$; each fiber is finite via polynomial fiber-finiteness (`Polynomial.finite_setOf_isRoot`) applied to the quadratic $As^2 + Bs + A/x$ obtained from the bridge `steffensen_step_p2_eq_sigma_p2_explicit`. Hence $\mathrm{vol}(\mathrm{Bad}_{x,\alpha}) = 0$.
- **★★★★★ Fully unconditional real McMullen at** $p = 2$ (Theorem 7.9): for every $x > 1$ and $\alpha = 1/\sqrt{x}$, `RealMcMullenP2`(x, α) holds with no hypothesis, axiom-clean.

11.2 Proven conditional on an explicit axiom

- Kung-Traub converse for $c = 2$ (Theorem 4.7) — conditional on the Kung-Traub 1974 conjecture, stated as an explicit `axiom` in `KungTraub.lean`.

11.3 Proven conditional on explicit hypotheses (no new axioms)

- Steffensen local attraction (Theorem 5.6) — conditional on a local quadratic bound $K \cdot r \leq 1/2$ on $B(\alpha, r)$ *(in $\mathbb{C}$; in $\mathbb{C}$ this hypothesis is now unconditionally discharged by Theorem 6.2).*
- Pointwise dynamical convergence from entry (Theorem 5.8) — same hypothesis, plus a first-entry time k_0.
- ★★★ Complex almost-everywhere dynamical convergence (Theorem 5.9) — conditional on (i) per-anchor local quadratic bounds (discharged ğ31) and (ii) a.e. trajectory entry (generic McMullen; stated as named Prop `McMullenAEEntry`).

Previously: `RealMcMullenP2`(x, α) at $p = 2$ was listed here, conditional on $\mathrm{vol}(\mathrm{Bad}_{x,\alpha}) = 0$. The Lebesgue-null hypothesis is now a formalised theorem (Corollary 7.8), and `RealMcMullenP2` has been promoted to the unconditional list (Theorem 7.9).

11.4 Not claimed

- An unconditional, fully Lean-certified a.e. trajectory-entry theorem in $\mathbb{C}$ at arbitrary p (the Fatou-type McMullen global dynamics). **The real-line $p = 2$ specialisation is now *fully* discharged axiom-clean with no residual hypothesis (Theorem 7.9, §7).**
- Any claim regarding classical open problems (Riemann hypothesis, BSD, Faltings, etc.). Previous speculative sections of this paper have been removed.

12 The `Pandrosion.Legacy` Companion Corpus

Alongside the `Core` spine, the development ships a 9-module companion under `lean/Pandrosion/Legacy/`. These modules are *not* load-bearing for any Core theorem — the dependency is strictly one-way, `Legacy` imports `Core` and not vice versa — but they formalise material that is routinely cited alongside the Pandrosion iteration: range invariants, the Aitken exactness theorem, explicit low-p contraction identities, a universal-in-p contraction bound, calculus facts, the no-cycle principle, the anchor-based cube-root variant, and the universal-descent theorem. The companion adds 44 theorems and 7 definitions to the build; every theorem passes the same axiom audit as the Core spine.

12.1 `Legacy.Basic` — range invariance and exact Aitken extrapolation

Theorem 12.1 (Range invariance, `h_lt_one` / `h_pos`). *For $x > 1$ and $p \geq 1$, $h_{p,x}$ maps $[0,1]$ into $(-\infty, 1)$; for $p \geq 2$, it maps $[0,1]$ into $(0,1)$. Consequently, every iterate $h^{[n]}_{p,x}(s_0)$ starting from $s_0 \in [0,1]$, $n \geq 1$, lies in $(0,1)$ (`orbit_in_interval`).*

Theorem 12.2 (Exact Aitken extrapolation, `aitken_perfect_extrapolation`). *Let $s, lam, r \in \mathbb{R}$ with $lam \neq 1$ and $s \neq r$, and set $s_1 := r + lam(s - r)$, $s_2 := r + lam(s_1 - r)$. Then the denominator $s_2 - 2s_1 + s = (s - r)(lam - 1)^2$ is non-zero, and*

$$s - \frac{(s_1 - s)^2}{s_2 - 2s_1 + s} \;=\; r$$

exactly*, in a single step. The Aitken Δ^2 accelerator is the identity recovery on any geometric error sequence with rate $lam \neq 1$.*

12.2 `Legacy.ContractionP2` — the $p = 2$ contraction identity

Theorem 12.3 (Explicit contraction identity, `contraction_identity_p2`). *For $x \neq 0$ and $1 + s, 1 + t \neq 0$,*

$$h_{2,x}(s) - h_{2,x}(t) \;=\; \frac{(x - 1)(s - t)}{x\,(1 + s)(1 + t)}.$$

Specialised to $t = s^\star = 1/\sqrt{x}$ this gives `distance_decreases_p2` (strict decrease) and `h_strict_mono_p2` (monotonicity); the combined monotone-from-above and monotone-from-below arguments yield `fixed_point_uni`

12.3 `Legacy.UniversalContraction` — universal divided difference

Definition 12.4 (Divided difference). $Q_p(s,t) \;:=\; \sum_{k=0}^{p-1}\sum_{j=0}^{k-1} s^j\, t^{k-1-j}.$

Theorem 12.5 (Universal contraction bound, `contraction_general`). *For every $p \geq 2$, every $x > 1$, every $s \geq 0$ and every fixed point $s^\star \in (0,1)$ with $(s^\star)^p = 1/x$,*

$$\big|h_{p,x}(s) - s^\star\big| \;\leq\; \frac{x - 1}{x}\,\big|s - s^\star\big|.$$

The proof factors $S_p(s) - S_p(s^\star) = (s - s^\star)Q_p(s, s^\star)$ and bounds $Q_p(s, s^\star) \leq S_p(s)\,S_p(s^\star)$ (`Qp_le_Sp_mul`) via an explicit monomial-counting argument. Concrete instantiations at $p = 3, 4, 5$ are proved directly from the factorisations `Sp3_sub`, `Sp4_sub`, `Sp5_sub`.

12.4 `Legacy.Derivative` — calculus of S_p and $h_{p,x}$

Theorem 12.6 (`HasDerivAt` for S_p, `Sp_hasDerivAt`). *For every $p \in \mathbb{N}$ and every $s \in \mathbb{R}$, S_p is differentiable at s with derivative $S'_p(s) = \sum_{k<p} k\, s^{k-1}$. For $p \geq 2$, $S'_p(s) > 0$ whenever $s \geq 0$ (`Sp'_pos`).*

Theorem 12.7 (`HasDerivAt` for $h_{p,x}$, `h_hasDerivAt`). *For $x > 0$, $p \geq 1$, $s \geq 0$,*

$$h'_{p,x}(s) \;=\; \frac{(x - 1)\,S'_p(s)}{x\,S_p(s)^2}.$$

In the square-root case $p = 2$ this simplifies to $h'_{2,x}(s) = (x - 1)/(x(1 + s)^2) \in (0,1)$ (`h_deriv_p2`, `asymptotic_rate_in_unit`).

12.5 `Legacy.NoCycles` — no periodic orbits

Theorem 12.8 (No periodic orbit under a contraction, `no_periodic_orbit`). *Let $f : \mathbb{R} \to \mathbb{R}$, $r \in \mathbb{R}$, $c \in [0, 1)$, and suppose $|f(s) - r| \le c\,|s - r|$ for every s. Then $f^{[n]}(s) = s$ for some $n \ge 1$ implies $s = r$.*

Specialised to $f = h_{p,x}$ via Theorem 12.5, this rules out every periodic orbit of the real Pandrosion iteration outside the fixed point — an elementary but crisp dynamical statement.

12.6 `Legacy.VoronoiAffine` — affine form of the Voronoï bisector

The coordinate-level statement underlying `Core.MultiStartBasins.halfPlane_convex`:

$$\|z - r_1\|^2 \;\le\; \|z - r_2\|^2 \iff 2\left(z_{\mathrm{Re}}(r_{2,\mathrm{Re}} - r_{1,\mathrm{Re}}) + z_{\mathrm{Im}}(r_{2,\mathrm{Im}} - r_{1,\mathrm{Im}})\right) \;\le\; \|r_2\|^2 - \|r_1\|^2.$$

The companion `multistart_distinct_roots_guarantee` complements `Core.fin_injective_iff_bijective`: on $\mathrm{Fin}\, d$, *any* surjection is already a bijection.

12.7 `Legacy.Descent` — universal descent $D_p < 0$

Theorem 12.9 (Universal descent, `normalized_descent_neg`). *For every $p \ge 2$,*

$$D_p \;:=\; \frac{1}{p}\sum_{k=1}^{p-1} \log\!\Big(\cos \tfrac{k\pi}{2p}\Big) \;<\; 0.$$

The proof is termwise: every angle $k\pi/(2p)$ with $1 \le k \le p-1$ lies in $(0, \pi/2)$ (`descent_angle_in_range`), on which cos takes values in $(0, 1)$ (`cos_in_unit_interval`), hence $\log\cos$ is strictly negative (`log_cos_neg`). The $k = 0$ term contributes $\log 1 = 0$, so the average is strictly negative.

12.8 `Legacy.CubeRoot` — cube-root specifics

Theorem 12.10 (Cube-root derivatives, `pandrosion_derivative_value` & `pandrosion_second_derivative_v` *Let $P(s) = s(s^3 + 4X)/(3s^3 + 2X)$. For every real cube root r of X (so $r^3 = X$, $r \neq 0$),*

$$P'(r) \;=\; -\tfrac{1}{5}, \qquad P''(r) \;=\; -\tfrac{12}{25\,r}.$$

In particular the Pandrosion cube-root iteration has a universal *linear rate $-1/5$ (independent of X), with Taylor expansion $P(r + \varepsilon) = r - \varepsilon/5 - 6\,\varepsilon^2/(25r) + O(\varepsilon^3)$. Since $-1/5 \neq 1$, Aitken-Steffensen applies; since $-1/5 \neq 0$, convergence is strictly linear (not quadratic, unlike Newton).*

Theorem 12.11 (Pandrosion is neither Newton nor Halley nor Chebyshev-Halley). *The polynomial cross-identities `pandrosion_newton_cross` and `pandrosion_halley_cross` yield $P - N \propto -(3s^3 - 2X)(s^3 - X)$ and $P - H \propto -s^4(s^3 - X)$ respectively: Pandrosion agrees with Newton and Halley only at the cube roots. Moreover, `pandrosion_not_in_chebyshev_halley` rules out the entire Chebyshev-Halley one-parameter family CH_α: matching the s^6 and s^3X coefficients forces $\alpha = 1/2$ and $\alpha = 2/5$ respectively, which are incompatible.*

12.9 `Legacy.AnchorStep` — anchor-based cube-root variant

For the cube-root problem $z^3 = X$, define the *anchor step*

$$F_a(s) \;=\; a - \frac{a^3 - X}{Q(a, s)}, \qquad Q(a, s) \;=\; s^2 + a\,s + a^2 \;=\; \frac{s^3 - a^3}{s - a}.$$

Theorem 12.12 (Anchor fixed-point, `anchor_fixed_point`). *For every anchor a and every cube root r of X with $Q(a, r) \neq 0$, $F_a(r) = r$.*

Theorem 12.13 (Newton as degenerate case, `newton_from_pandrosion`)**.** *For $a \neq 0$, the self-anchored step collapses to Newton: $F_a(a) = (2a^3 + X)/(3a^2)$.*

A complete multi-start step $(a, s) \mapsto (\widehat{a}, s')$ composes three anchor iterations with an Aitken reanchoring. The full pair is a fixed point at every cube root (`multistart_step_at_root`); `aitken_exact_geometric` gives the exact extrapolation form on a clean geometric orbit $(s, r + \lambda e, r + \lambda^2 e)$.

12.10 Module layout of `Pandrosion.Legacy`

Module	LOC	Headline content
`Legacy.Basic`	103	range invariance, orbit in $(0, 1)$, Aitken exactness
`Legacy.ContractionP2`	169	$p = 2$ contraction identity, monotonicity, uniqueness
`Legacy.UniversalContraction`	367	universal Q_p, $(x - 1)/x$ bound, $p = 3, 4, 5$
`Legacy.Derivative`	173	`HasDerivAt` for S_p, $h_{p,2}$; asymptotic rate
`Legacy.NoCycles`	74	no periodic orbit under a contraction
`Legacy.VoronoiAffine`	57	affine bisector form; surjective $\Rightarrow$ bijective on $\mathrm{Fin}\, d$
`Legacy.Descent`	122	$D_p < 0$ via $\log\cos$ bounds
`Legacy.CubeRoot`	173	cube-root $P'(r) = -1/5$, $P''(r)$, Chebyshev-Halley exclusion
`Legacy.AnchorStep`	223	anchor step F_a, multistart, Newton as $a = s$

13 Reproducibility and Tooling

The complete Lean source lives in `lean/Pandrosion/Core/*.lean` (24 files) and `lean/Pandrosion/Legacy/*.le` (9 files), with the aggregator `Pandrosion/Legacy.lean` and the root module `Pandrosion.lean` importing both layers. The development targets Lean 4 with Mathlib v4.7.0 and is built with a Docker-based incremental pipeline:

```
docker compose run --rm lean-incremental
```

Listing 3: Incremental build command

On a clean tree, the full 34-module corpus compiles successfully (`296` theorems, `59` definitions, 0 `sorry`, 0 off-whitelist axioms); on an incremental rebuild after a single-file edit, only the changed module and its downstream dependents are rebuilt. The stricter service `docker compose run -rm lean-check` reruns the axiom audit on every declaration against the whitelist {`propext`, `Classical.choice`, `Quot`

The source code is publicly available at github.com/ivan-fr/poussiere_de_x.

14 Conclusion

This paper documents a Lean 4 / Mathlib v4.7.0 development of the Pandrosion-Steffensen algorithm for $z^p = x$, consisting of a 24-module `Pandrosion.Core` spine and a 9-module `Pandrosion.Legacy` companion. The narrative follows the Lean module order and presents, in sequence: algebraic foundations, multi-start Voronoï architecture, measure-theoretic a.e. coverage, cyclotomic anchor family, closed-form linear contraction rate, p-uniform complexity, Kung-Traub efficiency optimality, complex multiplier in closed form, conditional local attraction, the crown-jewel a.e. dynamical convergence theorem (mod McMullen), a *fully unconditional* axiom-clean proof of the sharpened real-line McMullen statement at $p = 2$, $x > 1$ (Theorem 7.9), and — as the universal capstone of the corpus — the ğğ93–95 *multi-start universal loglog convergence theorem* (Theorem 9.4), which gives, for every $x \in \mathbb{C}^*$, every $p \geq 1$, every cyclotomic anchor α, every $z \in \mathbb{C}$ and every precision $\varepsilon > 0$, an explicit perturbative seed $\alpha + \varepsilon_{\text{seed}} \cdot (z - \alpha)$ and an iteration count $N = O(\log\log(1/\varepsilon))$ such that the Steffensen iterate from that seed is within ε of α at every step beyond N.

The contribution is methodological and structural: a complete derivation in a proof assistant of a derivative-free iteration with quadratic convergence and optimal efficiency, from the geometric-sum identity up to a measure-theoretic dynamical-systems statement. Where genuinely analytic content remains deferred at full generality (a.e. trajectory entry in $\mathbb{C}$ at arbitrary p), the hypotheses are explicit and auditable; the real-line $p = 2$ specialisation, by contrast, is now fully and unconditionally discharged, with the bad-set Lebesgue-null property itself proven via polynomial fiber-finiteness. The ğğ93–95 multi-start universal theorem closes the algorithmic gap left by the a.e. statements: it gives an everywhere-convergent computational procedure with prescribed-root targeting and optimal loglog complexity, three properties not achieved simultaneously by any classical algorithm (Newton, Halley, plain Steffensen) in the literature.

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